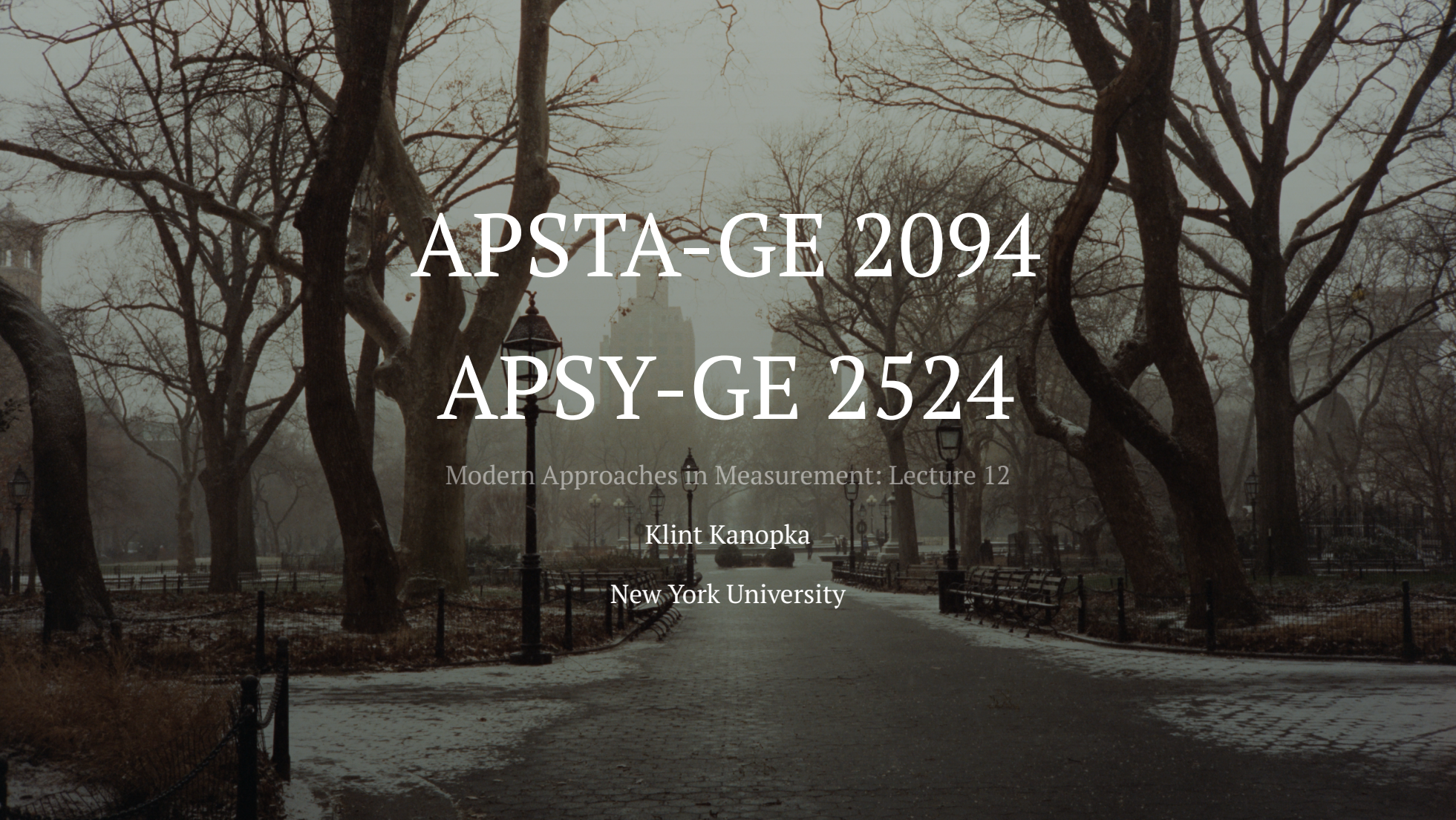

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Modern Approaches in Measurement: Lecture 12

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Homework Stuff

- For PS5:
  - Aim to have more than 250 complete cases for your analysis
  - Only use the survey questions in your factor analysis and LCA
  - Remember that FA makes groups of items and LCA makes groups of respondents
  - Use the demographic questions you pick to help describe your latent classes
  - The GSS racial categories are clunky; feel free to transform them into something more useful
- Another note: As we wind down, think about any specific topics or applications you're interested in or might want to learn more about

Recommendations

Recommendations

- How do you learn about new stuff?
  - Recommendations!
  - Broadly we can think about two types of recommendations:
    - Generic
    - Personalized
- Generic recommendations don’t care about who you are
  - Wirecutter articles
  - Curated lists from influencers
  - Best selling
  - Most popular
  - Trending
  - Recent uploads
- Personalized recommendations are, well, *personalized*
  - Suggestions from friends

Formulating the Problem

- Recommendations require three parts:
  - X is the set of *users* (customers, viewers, etc.)
  - S is the set of *items* (products, posts, videos, songs, etc.)
  - The *utility function* $u : X \times S \to R$
    - R is a set of ratings
    - R is *totally ordered*, meaning any two elements can be compared
      - During comparisons, one item is either rated higher than the other or they are equally rated
      - Stars, thumbs up/down, swiping left/right, etc.

The Utility Matrix

- Ratings are stored in an object called the *utility matrix*
- The problem is that every user doesn't rate every item
  - The utility matrix is *sparse*

	Taylor Swift	Bad Bunny	Turnstile	Pantera	Crowbar
Alice	5	3	3		
Bob	1			5	
Carol	4		5		
David			4	4	5

- Does this remind you of anything we've dealt with before?

The Two Core Problems of Recommendation

- Problem One: Gathering observed ratings
  - This is the data collection problem
  - Where might this information come from?
    - Explicit ratings
      - Ask people to rate items
      - Pay people to rate items
    - Implicit ratings
      - Learn ratings from user behavior
- Problem Two: Predicting unobserved ratings
  - Upside is that you only really care about predicting high unknown ratings!
  - Downside is that this is a tricky problem and measuring success isn't always straightforward
  - Any early ideas on how this can be done?

Movie Recommendation

- With a group of 3 ± 1:

1. Download the MovieLens 100K Dataset
2. Unzip the files, check out the `README` to get a sense of what's in there and how the data are organized
3. The full dataset is in `u.data`. Using whatever you like from the files provided, come up with a plan for recommending new movies to users
4. Try to code up a couple of test cases or mild proof-of-concepts
5. Start with something *very* simple and then add complexity!

Approach One: Content-based Recommendations

- **Core idea:** Items have features and you recommend new items to a user that have features similar to items they've already rated highly
  - Restaurants that have the same owner or head chef
    - If you like Thai Diner, you should try Mommy Pai's
  - Bands that share members
    - If you like Pantera and Crowbar, you should listen to Down
  - Movies by the same director
    - If you like Raiders of the Lost Ark and Hook, you should watch Schindler's List
- How do you decide on the features?
  - Hand pick them
- How do you decide on similarity?
  - Use a distance metric!

Approach One: Content-based Recommendations

- Pros:
  - Independent of other users
  - You can handle users with niche tastes
  - You can recommend new and unpopular items
  - Recommendations are explainable
- Cons:
  - Feature engineering is hard
  - Recommending to new users doesn’t really work
  - Limits the diversity of things you show to a user

Approach Two: Collaborative Filtering

- **Core idea:** We find users who have rated things similarly to you and then recommend to you things they've liked
  - We use the observed ratings of an item as its feature set
  - We use the observed ratings of a user as its feature set
  - Recommendations are made based on matching similar users (user-user collaborative filtering) or matching similar items (item-item collaborative filtering)
- How might this work in practice?

User-User Collaborative Filtering

- For a user $x \in X$:
  - Find the set of K other users who have ratings most similar to x
  - Estimate x's ratings on unseen items using the ratings of those K users
- How might you determine similarity of ratings?
  - Cosine similarity, Jaccard similarity, correlations, lots of options!
- How would you then estimate x's rating on a new item?
  - You can take the average of other users' ratings
  - You can take a weighted average using their similarity to x

Item-Item Collaborative Filtering

- For a user x and item $s \in S$:
  - Find the set of K other items who have ratings most similar to x
  - Estimate x's rating of s using the ratings of those K items
- This generally boils down to k-Nearest Neighbors (kNN) followed by a weighted average
- Which version do you think works better? Why?
  - Item-item usually works better, as items are simpler

Approach Two: Collaborative Filtering

- Pros:
  - Works for any kinds of items
  - Does not require feature selection
- Cons:
  - Cold start problems
    - You need a critical mass of users and ratings
  - Sparsity problem
    - Might be hard to find users who have rated similar items
    - Unable to recommend items that have never been rated
  - Popularity bias
    - Users with unique tastes don’t get good recommendations
    - Tends to over recommend popular items

Dating Apps

Dating Apps

- I think most people have tried dating apps?
- A rough dating app typology:
  - Question-based
    - OKCupid and older apps
  - Swipe-based
    - Tinder, Hinge, etc
  - Location-based
    - Grindr, Scruff
- Other stuff?

Dating Apps

- What's the goal of a dating app?
  - Make money
    - Sell premium features
    - Maximize ad exposure
  - Connect people
    - If people don't like who they see, they won't use the app
    - If nobody talks to a person, they won't use the app
- What could we do if we design our own app?
  - How would you design the app?
  - How would you match users?
  - Talk to a few neighbors about it!

Tinder

- Tinder used to use an Elo system (Bumble, too)
- The probability person i swipes right on person j, $X_{ij} = 1$, depends on the difference between their- latent scores in the app
  - Assumes people will swipe left $(X_{ij} = 0)$ on people they view as "beneath them"
  - Goal: Display profiles to maximize the probability that they swipe right *on each other*
  - When is this probability maximized?
- Turns out this made people really upset
- Now they display people based on app usage

Hinge

- Hinge is “designed to be deleted” or something
- They treat dating as a stable matching problem
  - Assume we have two groups that attempt to match with each other, A, B
    - Pairings are not stable if a member of A prefers some member of $B over the one they’re matched with
    - *and* the member of B they prefer also prefers them over who the member of A they’re matched with
- The solution to this is the Gale-Shapley Agorithm (which won the 2012 Nobel Prize in Economics)
  - This requires group members to develop rankings of members of the other group

Gale-Shapley Algorithm

- How it works:
  - In each iteration, members of group A who have not yet matched make an offer to their top choice in group B who they have not yet made an offer to
  - Members of group B evaluate their offers against their current match and either accept the best new offer or reject all offers
  - Repeat until all members are matched
- Guaranteed to match everyone
- Guaranteed to produce stable matches

Back to Hinge

- What's the problem?
- Users don't produce rankings of everyone they're trying to date!
- How does Hinge infer rankings?
  - They use interaction patterns to determine the type of users you prefer
  - They're not specific about the methods they use, but we could make some guesses!
  - How would you do it?

OKCupid

- Individuals respond to questions and describe how their ideal partner would respond to those questions
  - Additionally they specify how important responses are to them
- OKCupid provides users with “Match %”
  - They’re not super specific on what the actual model is
- How could we do this?

OKCupid Experiments

- OKCupid has been really open about the fact that they collect lots of data and experiment on their users
- The book *Dataclysm* is a super interesting documentation of applied statistics, measurement, and data science from the founder of OKCupid
- In 2013 OKCupid made a bunch of people pretty upset:
  - They basically just randomly rearranged the match scores
  - Turns out, if you tell people they're a high percentage match, they're more likely to talk to each other and keep talking
- The OKCupid Data Blog used to be full of cool stuff, but it's really only accessible via the Wayback Machine

Wrap Up

Recap

- We can build mixture models out of many distributions, including item response functions
- CDMs combine latent class analysis with theory to model categorical latent skill profiles
- In CDMs, class membership is defined by which attributes a respondent possesses
- Process data are collected while respondents interact with items
- Response time is the most common form of process data and can be useful if we have a clear plan for using it